

2. Computer System ka Abstract View aur Component Hierarchy

Computer Science Fundamentals | Module 1: OS Architecture | CS_Fundamental-00

Reference to Previous Concept: Jaise humne **Concept 1.4** mein dekha ki computer system ke 4 components hote hain (Hardware, OS, Application Programs, Users), yeh topic **Image 2** par aadharit hai jo in components ka layered **Abstract View** aur right side mein **Translators** ka flow explain karta hai.

2.1 Abstract View of Components (Layered Architecture)

- **Layer 1 (Bottom Most) - Computer Hardware (H/W):** Core machine jisme CPU, Main Memory (RAM), aur I/O devices aate hain.
- **Layer 2 - Operating System (OS):** Hardware ke theek upar rehta hai aur hardware ko wrap karke safe, uniform services deta hai.
- **Layer 3 - System & Application Programs:** OS ke upar different utilities run hoti hain: COMPILER, ASSEMBLER, TEXT EDITOR, DATABASES, SPREADSHEETS, GAMES.
- **Layer 4 (Top Most) - Multiple Users:** User 1, User 2, ... User N jo concurrently system use kar rahe hote hain.

2.2 Translators (Handwritten Note Reference from Image 2)

Image 2 ke top-right corner mein likha hai: TRANSLATORS :- 1) ASSEMBLER 2) COMPILER, INTERPRETER | HLL -> 0, 1.

- **Translators ki Zaroorat:** Computer Hardware (CPU) sirf aur sirf **Machine Language (0 aur 1 / Binary Bits)** samajhta hai. Hum insaan high-level English-like code likhte hain (C, C++, Java, Python). Translators ka kaam iss Human-readable code ko Computer-readable binary bits (0, 1) mein convert karna hai.
- **2.2.1 Assembler:** Assembly Language (Low-Level Language jaise MOV AX, BX) ko seedhe Machine Language (0 aur 1) mein convert karta hai.
- **2.2.2 Compiler:** High-Level Language (HLL jaise C/C++) ke poore code ko ek saath compile karke machine code generate karta hai.
- **2.2.3 Interpreter:** High-Level Language (jaise Python, JS) ko line-by-line interpret karke execute karta hai.

Fig 2.1: Abstract View of Computer System Components

Reference: Image 2 [Layered Components & Translators Architecture]

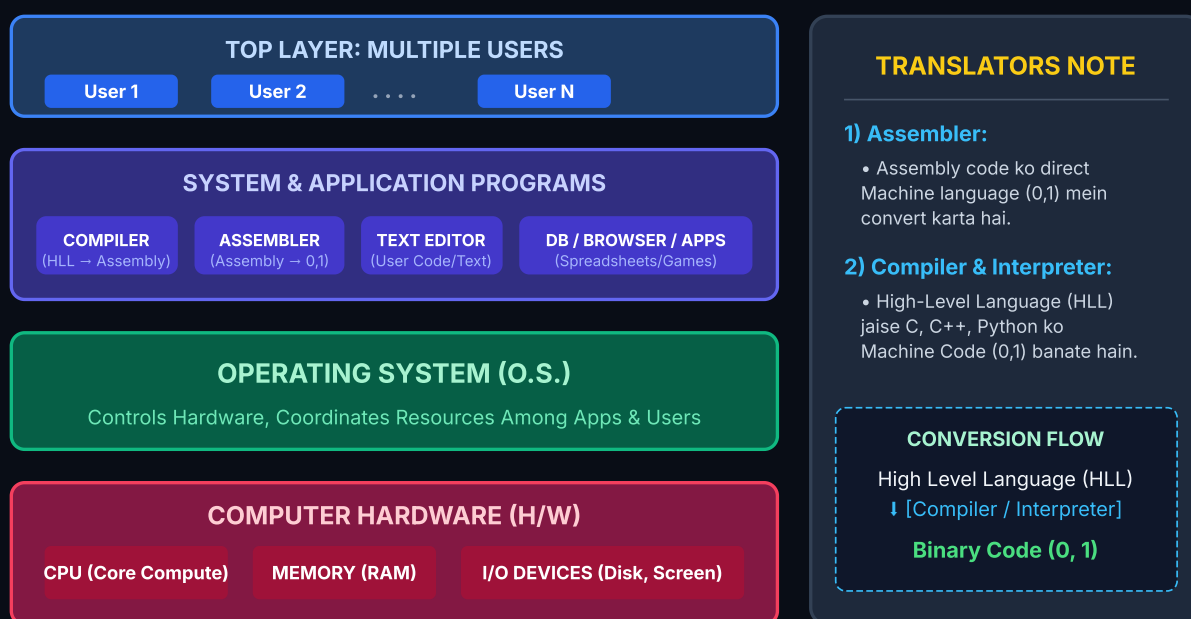


Fig 2.1: Abstract View of Computer System & Translators Flow

2.3 Hardware Resources (H/W Deep Dive)

- **(a) The Memory (RAM):** Programs ko execute hone ke liye memory chahiye hoti hai. Variables aur active data yahin store hota hai.
- **(b) Input & Output (I/O) Devices:** Disks (Storage), Keyboard, Mouse, Monitor, Network cards jinke zariye data in aur out hota hai.
- **(c) CPU (Central Processing Unit):** Basic computing resources aur calculations execute karne wala brain.

2.4 Application Programs vs OS Coordination

Application programs (Word, Spreadsheets, Games, DBMS) solve user computing problems. OS in programs ke beech hardware resources ka coordination karta hai taaki koi application hardware ko monopolize na kare aur system stable bana rahe.